

Ilyan Radjabaly

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FORMATION

ENSAE Paris – Polytechnic Institute of Paris

Palaiseau, France

2nd Year Statistician Engineering Cycle (GEI competition ranked 20th out of 620) - 3.9 / 4.0 (15.13 / 20)

2025 – 2028

- **Courses :** Machine learning, Statistical learning, Data science, Algorithms, Databases, Stochastic processes, Measure theory and Lebesgue integrals, Topology and Hilbertian analysis, Bilinear algebra, Differential calculus, Differentiable optimization

Paris Dauphine University - PSL

Paris, France

Bachelor's Degree in Economics & Financial Engineering - 15.2 / 20

2022 – 2025

- **Ranking :** 5 / 620 ; 19 / 633, Top 1%
- **Computer science :** Programming (Python, SQL), Algorithms, Financial Programming (VBA) & Statistics (R, STATA, GRETL)
- **Economics :** Microeconomics, Public Economics, Macroeconomics (IS-LM, Phillips curve), Industrial Economics, Econometrics
- **Finance :** Insurance, Portfolio Management, Technical Analysis, Accounting, Financial Mathematics, Products & Markets
- **Mathematics :** Analysis, Multivariable Calculus, Static Optimization, Linear Algebra, Ordinary Differential Equations, Statistical Inference 1, Probability (Theory of Point Estimators), Statistical Inference 2 (Hypothesis Testing, ANOVA)
- **Social Sciences :** Public Law, Private Law, European Law, Contract Law, Sociology & Political Science, Management, Philosophy

Boston College

Chestnut Hill, MA, USA

Exchange semester- 3.75 / 4.0

August 2024 – December 2024

- **Courses :** Applied econometrics + Visualization and modeling with STATA (A-), Mathematical statistics + Statistical programming in R (A), Mobile programming in Swift/Swift UI (B+), Financial accounting (US GAAP) (A)

EXPERIENCE

AG2R La Mondiale - Capital Management

Paris, France

Quantitative Researcher

June 2026 – September 2026

- Statistical modeling of extreme scenarios in the context of Reverse Stress Tests: mathematical formalization of shocks.
- Stochastic Monte Carlo simulations (Solvency II): parameterization, execution in Python and analysis of balance sheet distortion
- Estimation and allocation of risk budgets from model outputs, via statistical analysis of simulated distributions (ALM)
- Mathematically justified definition of operational limits and construction of statistical indicators (KRIs) for risk monitoring
- **Project:** modeling the distribution of returns of the MSCI World in order to reproduce an extreme quantile (99.5%, -39%); comparison of parametric approaches vs nonparametric, study of thick tails and application of the kernel method.
- **Additional work:** optimization of the pseudo-random generator and improvement of the reliability of simulation procedures.

J-PAL Europe Lab of MIT, École Normale Supérieure & Paris School of Economics

Paris, France

Research Assistant – Data Collection

April 2025 – July 2025

- **Research objective:** Evaluate the impact of an AI-driven platform on students' reading skills, focusing on the application of Randomized Controlled Trials (RCTs, E. Duflo) in the economic field.
- Data collection and processing for development economics and econometrics projects; quantitative data analysis for academic publications; contribution to studies on economic policies and their impact

PolyExpert Insurance Group

Paris, France

Insurance Risk Expert Assistant

January 2025 – April 2025

- Insuredmanaging insurance files for the consulting group, using analytical methods for clients such as Groupama and AXA, through performance-risk analyses.

COMPANYSARL – Soap Factory “Savon SEIM”

Antananarivo, Madagascar

Data scientist

June 2024 – August 2024

- Developed and implemented a sales analytics dashboard enabling managers to track key performance indicators (KPIs) and identify sales trends. Designed and deployed an inventory tracking tool using company sales data, helping managers more effectively monitor stock levels and plan replenishment.

PUBLICATIONS & PROJECTS

- **Working paper in macroeconometrics :** Test the predictive effectiveness of the IS-LM model in a closed economy using multiple regression (STATA and R) and study the impact of the policy mix on US GDP; R-2 of 98% and a 3% forecast/actual difference
- **Ongoing statistical research project :** “Astrophysical Insights into Black Hole Mass Distribution : A Computational Statistic Inference Approach via Stochastic Modeling under the Likelihood Principle”
- **Working paper en econometrician.i.e. time series :** “The Impact of Internet-Driven Productivity Acceleration on Labor’s Marginal Contribution to GDP Growth through the Solow-Swan Model”
- **Ongoing applied mathematics research project :** “Study of the stability of continuous neural networks using systems of differential equations: convergence by qualitative analysis and implications for machine learning”
- **iOS app developmentSocialSquad :** social network with MVVM architecture withSwiftUI et Firebase (Cloud Firestore).

SKILLS AND CERTIFICATIONS

- **Languages :** French C2, English C1 (TOEFL iBT: 101/120, Cambridge First: Grade A, 180) and Malagasy C2
- **Programming :** Python (NumPy, Panda), VBA, SQL, GRETL, Swift & SwiftUI, R, STATA, Firebase & Cloud Firestore, LaTeX
- **Activities :** Member of the MUN association in Antananarivo; Freelance online video editor for 5 years, +50 commissions
- **Prizes :** Paris-Saclay Agglomeration Prize in the “Saclay 20” startup idea competition for our “MySaclay” app concept. Shortlisted to participate in the General Economics Competition and the General Mathematics Competition, as well as the 2021 Math Olympiad.